Measurement Report

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Abstract: Unbalanced Mach-Zehnder interferometers (MZIs) have been widely used to extract fabrication parameters; from their free-spectral-range (FSR) and spectral response, one can extract group index, dispersion, and even infer geometric parameters of the guiding core. This design proposal puts these ideas into practice with a family of custom MZI variants implemented in Luceda IPKISS using a custom-made PCell. The PCell instantiates directional couplers and fiber grating couples from the textscSIEPICPDK PDK, and realizes the path-length difference (Delta L) with spirals in the long arm, chosen over a serpentine to achieve larger delays within a smaller footprint, while respecting bend-radius and spacing rules. The proposed workflow provides a practical path to group-index and loss estimation using interferometric structures.

Index Terms: group index, waveguide losses, interferometers, silicon photonics.

1. Introduction

Unbalanced Mach–Zehnder interferometers (MZIs) have been widely used to extract fabrication parameters; from their free-spectral-range (FSR) and spectral response, one can extract group index, dispersion, and even infer geometric parameters of the guiding core. Moreover, by pairing "narrow" and "widened" straight sections, MZI variants can decouple height–width variations in actual fabricated waveguides [1]. At the wafer scale, using appropriate interferometer orders, i.e., selecting the FSR or the path-length difference ΔL, enables unambiguous mapping of effective index and loss across full wafers [2].

Beyond pure spectral analysis, the same unbalanced MZI measured with Integrated Optical Frequency-Domain Interferometry (I-OFDI) acts as a power-splitter test device (1-PSTD), allowing splitter-ratio and waveguide-loss extraction directly from the transmission response [3], [4]. For robust I-OFDI, a medium-to-long ΔL is desirable so that (i) the accumulated loss in the long arm is resolvable given the amplitude resolution of the system and (ii) the two main time-domain features are separated beyond the instrument's temporal resolution [4].

This design proposal puts these ideas into practice with a family of custom MZIs implemented in LUCEDA IPKISS. Three variants targeting different ΔL's were laid out to extract from their interference pattern the group index of the fabricated devices. Simulation results in this report use the actual layouts converted to circuit models for S-parameter analysis in IPKISS. In a second phase, these same devices are compatible with I-OFDI/1-PSTD measurements to extract splitter ratio and propagation loss without additional external interferometric hardware [3], [4]. Together, the proposed layouts and workflow provide a practical path to group-index and loss estimation

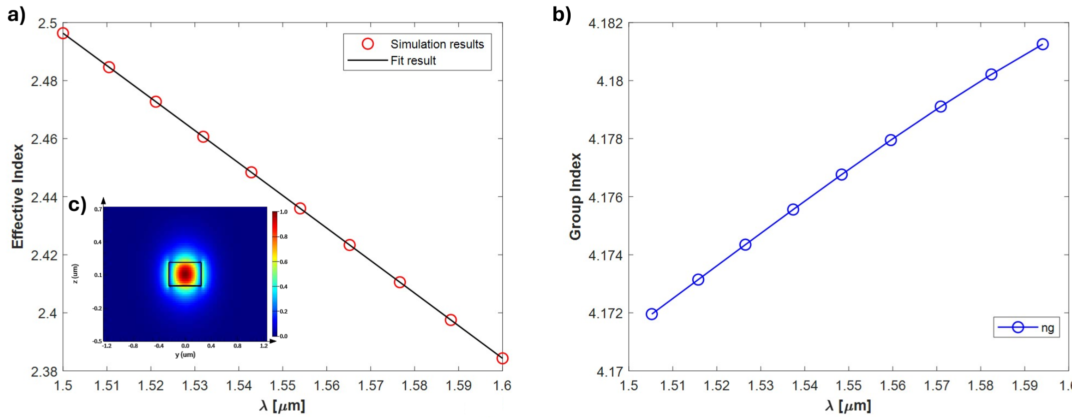

Fig. 1. (a) SOI $550\ \mathrm{nm} \times 220\ \mathrm{nm}$ waveguide TE0 effective index obtained using LUMERICAL MODE (red), with the second degree polynomial fit of the results for the waveguide compact model (black) and (b) group index between $1.5\ \mu\mathrm{m}$ and $1.6\ \mu\mathrm{m}$. (c) TE0 mode profile obtained with LUMERICAL MODE

using interferometric structures.

2. Modeling and Simulation

2.1. Waveguide modeling: Effective index, group index and compact model.

Figure 1 summarizes the workflow for modeling and results for the TE_0 mode of a silicon on-insulator (SOI) rib waveguide with cross section $550 \times 220\,\mathrm{nm}^2$. The frequency sweep in $\lambda \in [1.50,\ 1.60]\,\mu\mathrm{m}$ was performed in LUMERICAL MODE using a script provided by the course, which automates the extraction of the eigenmode over the wavelength and the exports $n_{\mathrm{eff}}(\lambda)$. The script was configured with the grid parameters obtained from the grid-convergence tests carried out for the course, ensuring that both n_{eff} and the TE_0 field profile (Figure 1 (c)) are numerically stable with respect to mesh refinement. Because all subsequent designs and measurements target the TE polarization, only the fundamental TE_0 mode is considered here.

From the sampled $n_{\mathrm{eff}}(\lambda)$ data (Figure 1 (a), red), a quadratic polynomial fit was obtained from the simulation data, yielding in the so-called waveguide 'compact model':

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2\,(\lambda - \lambda_0) + n_3\,(\lambda - \lambda_0)^2. \tag{1}$$

The black curve in Figure 1 (a) shows agreement between the fitted model and the MODE samples over 1.50–$1.60\,\mu\mathrm{m}$. The group index in Figure 1 (b) is then obtained consistently from the compact model through $n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda\,\mathrm{d}n_{\mathrm{eff}}/\mathrm{d}\lambda$, resulting in smooth $n_g(\lambda)$ suitable for use in interferometric predictions (e.g., $\mathrm{FSR}(\lambda)$). The fitted coefficients are as follows:

$$n_1 = 2.4435, \qquad n_2 = -1.1306, \qquad n_3 = -0.0418.$$

Evaluated at the reference wavelength, the compact model gives a

$$n_g(\lambda_0) = n_1 - \lambda_0\,n_2,$$

which with the above coefficients yields $n_g(@1.55\,\mu\mathrm{m}) \approx 4.1959$.

2.2. Mach-Zehnder Interferometer

Short, medium and long path-length differences (ΔL) were selected for this design to ensure the devices also operate as 1-PSTD structures. Three long path-length differences were chosen:

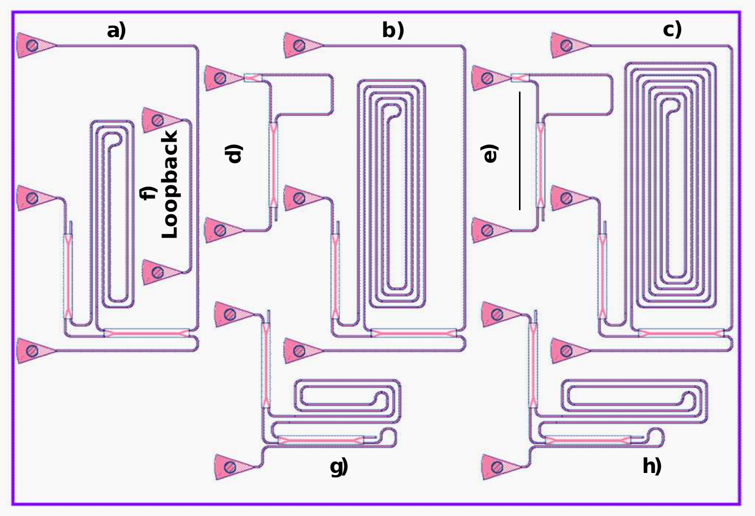

Fig. 2. (**a**–**c**) Seven design variants of an unbalanced Mach–Zehnder interferometer (MZI) implemented in Luceda IPKISS using a custom-made PCell. The devices were built using directional couplers and grating couplers of the SIEPIC PDK; only the long-arm path-length difference varies: (**a**) $\Delta L = 1000\,\mu\mathrm{m}$, (**b**) $\Delta L = 2000\,\mu\mathrm{m}$, (**c**) $\Delta L = 3000\,\mu\mathrm{m}$, (**d**) $\Delta L = 100\,\mu\mathrm{m}$, (**e**) $\Delta L = 120\,\mu\mathrm{m}$, (**f**) Loopback reference used to normalize the MZI responses by removing the contribution of grating couplers.(**a**) $\Delta L = 600\,\mu\mathrm{m}$, and (**b**) $\Delta L = 800\,\mu\mathrm{m}$.

$\Delta L = 1000, 2000, 3000\,\mu\mathrm{m}$, two medium $\Delta L = 600, 800\mu\mathrm{m}$ and two short $\Delta L = 100, 120\,\mu\mathrm{m}$. A custom PCell in LUCEDA IPKISS was implemented to achieve any other ΔL. By adjusting such parameter it is possible to automatically generate the layout and simulation of each cell.

Equation (2) gives the free-spectral range (FSR) for the MZI:

$$\mathrm{FSR} \approx \frac{\lambda^2}{n_g\,\Delta L} \tag{2}$$

For the target ΔL values, this yields approximately 0.572 nm, 0.286 nm, 0.191 nm,0.954 nm, 0.716 nm, 5.726 nm and 4.771 nm respectively, which are well within the resolution provided by the 1 pm step size of the tunable laser (TLS) described in the course materials. Note that there is a huge step between the short and medium PLDs (100 to 600 μm) which corresponds to a code error when using the layout tool. The initial intention was to implement a 300 μm delay, but the vector value was left on 120 μm.

The main challenge with the selected path-length differences is the on-chip footprint required to implement them. One option is to realize the delay with spirals, leveraging the small minimum bend radius achievable in silicon-on-insulator (SOI) technology; this approach minimizes area. However, to enable the extraction of waveguide geometric parameters as described in [1], the path-length difference should be implemented primarily in straight sections. We therefore consider spirals with a large bend radius and long straight sections.

2.3. Mach–Zehnder Interferometer layout and simulation using LUCEDA IPKISS

All layouts were generated in LUCEDA IPKISS with a custom-made PCell implementing an unbalanced MZI. The PCell instantiates directional couplers and fiber grating couplers from the SIEPIC PDK, and realizes the path-length difference ΔL with a spiral in the long arm, chosen over a serpentine to achieve larger delays within a smaller footprint, while respecting bend-radius

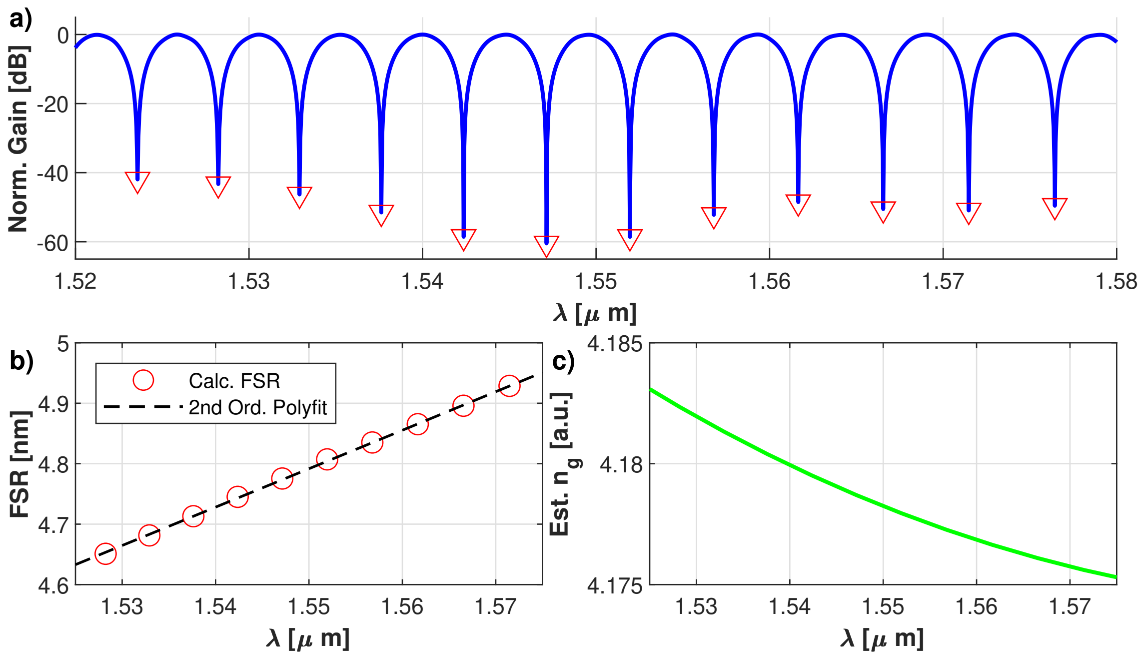

Fig. 3. (**a**) Simulated gain spectrum of the unbalanced MZI with $\Delta L = 120\,\mu\mathrm{m}$, implemented in Luceda IPKISS and post-processed to remove the grating-coupler response. (**b**) Free-spectral range (FSR) as a function of λ, extracted from the notch spacing in (a); a second-degree polynomial fit (black dotted curve) is applied to mitigate peak-position uncertainty due to the wavelength resolution. (**c**) Estimated group index obtained from the fitted FSR in (b) using Eq. (2).

and spacing rules. Seven MZI variants targeting $\Delta L = 100\,\mu\mathrm{m}, , 120\,\mu\mathrm{m}, , 600\,\mu\mathrm{m}, , 800\,\mu\mathrm{m}$, $1000\,\mu\mathrm{m}$, $2000\,\mu\mathrm{m}$, and $3000\,\mu\mathrm{m}$ were created. To hit each target precisely, the straight section of the spiral was optimized with `scipy.optimize.minimize`, which adjusts the trim length given the discrete geometry of turns, radius, and pitch. The realized delays had a maximum of $\approx 0.03\%$ deviation from the target values. A loopback reference was included on the same die to normalize MZI measurements by removing the grating-coupler contribution. The final mask was exported to GDSII directly from IPKISS (Figure 2).

Simulations of the *actual* device layouts were implemented to check the circuits behavior prior to fabrication. Figure 3 shows the results for one design variant ($\Delta L = 120\,\mu\mathrm{m}$) to illustrate the data analysis flow implemented. Each variant was converted in IPKISS to a circuit model for S-parameter analysis, and the transmission parameter $|S_{21}|$ was obtained (Figure 3 (a)). The spectra were then normalized with the loop-back measurement to remove the wavelength-dependent response of the grating couplers. The free-spectral range $\mathrm{FSR}(\lambda)$ was computed by a peak/notch search on the normalized spectra; because the wavelength sampling resolution can bias local FSR estimates at these small spacings, a second-degree polynomial was fitted to $\mathrm{FSR}(\lambda)$ (Figure 3 (b)), consistent with Eq. (2). Finally, the fitted $\mathrm{FSR}(\lambda)$ was used in Eq. (2) to recover the group index $n_g(\lambda) = \lambda^2/(\mathrm{FSR}_\lambda\,\Delta L)$, as shown in Figure 3 (c).

Simulation results after processing for the seven design variants proposed for this design are presented in Table I. Comparing with the target $n_g(@1.55\,\mu\mathrm{m}) \approx 4.1959$, it becomes clear that with the proposed data processing flow, the estimated n_g will be closer to the theoretical value for larger delays, mainly due to the uncertainty on the FSR polynomial fit, that with short PLDs will have less points.

TABLE I
SIMULATED FSR AND GROUP INDEX FOR DIFFERENT ΔL.

ΔL [μm]	FSR [nm]	n_g est.
100	5.780	4.176
120	4.807	4.179
600	0.956	4.193
800	0.717	4.193
1000	0.574	4.194
2000	0.287	4.195
3000	0.191	4.195

3. Fabrication

3.1. Fabrication details

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery [5].

3.2. Corner Analysis

To assess sensitivity to fabrication tolerances, four corner geometries were evaluated by shifting the width to 470 nm or 510 nm and the height to 215.3 nm or 223.1 nm. The range of waveguide widths considered in this corner analysis is based on measured variations from fabricated devices, while the range of heights corresponds to the silicon thickness variations reported by the foundry.

The results shown in Fig. 4 were obtained from *Lumerical MODE*, where each waveguide geometry was simulated. Figure 4 shows the simulation results for the effective index n_{eff} and the group index n_g of the waveguide for variations in its width and height. The nominal geometry $(500\ \mathrm{nm}, 220\ \mathrm{nm})$ is represented by the purple marker and yields $n_{\mathrm{eff}} = 2.4435$ and $n_g = 4.1954$.

The resulting indices are summarized in Figure 4. Across the fabrication corners considered, the experimental group index extracted from the measurements should be in the range $n_g \in [4.1606,\ 4.2398]$. In the figure, the highest value of n_g is highlighted in green and the lowest value in red.

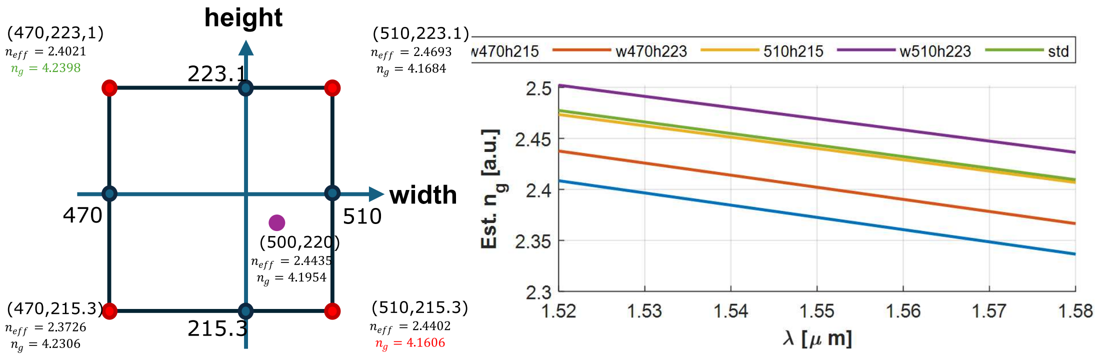

Fig. 4. Corner analysis of the waveguide eigenmodes obtained with Lumerical MODE, using width variations measured from fabricated devices and height variations provided by the foundry. The nominal geometry and four fabrication corners are shown, together with the corresponding $n_{\rm eff}$ and n_g values. The highest n_g is marked in green and the lowest in red.

4. Experimental Data

4.1. Measurement Description

To characterize the devices, a custom-built automated test setup [6], [7] with automated control software written in Python was used. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [8]. A 90º rotation was used to inject light into the TM grating couplers [8]. A polarization maintaining fibre array was used to couple light in/out of the chip [9].

4.2. Measurement Results

The raw transmission data obtained from the tunable-laser sweep were first imported into the data-processing pipeline. Each measurement consisted of two traces: a loopback reference and the corresponding unbalanced MZI. The loopback captured the wavelength-dependent response of the device under test.

The processing workflow was as follows. First, the raw loopback trace was smoothed to obtain its spectral envelope. This envelope was then used to normalize the MZI spectra by removing the baseline contribution of the grating couplers, leaving only the interferometric response of the unbalanced MZI. Once normalized, the signal was suitable for peak and notch detection, enabling extraction of the free-spectral range (FSR) required for subsequent estimation of the effective and group indices. Figure 5 presents the latter pipeline applied to the $\Delta L = 120\ \mu m$ case.

5. Analysis

The processed MZI traces were analyzed using the method introduced in the course materials. Starting from the distance between adjacent minima in the normalized spectrum, an initial estimate of the free-spectral range FSR(λ) was obtained. Using the known path-length difference ΔL, an initial estimate of the group index was computed. This value was then used as the starting point for the nonlinear fit algorithm of the MZI transfer function, from which the final estimates of n_g and $n_{\rm eff}$ were obtained.

Figure 6 shows the full workflow applied to the $\Delta L = 120\ \mu$m case: (a) the measured normalized transmission, (b) the measured FSR and group index as a function of wavelength, (c) the fitted transfer function over the experimental data, and (d) the retrieved group index function and the value at λ_0. For this device, the result was

$$n_g(1.55\ \mu\text{m}) = 4.1947.$$

The same algorithm was applied to the seven design variations, yielding in the results shown in Table 5; comparing the simulated and measured FSR and group index values for ΔL ranging from 100 μm to 3000 μm. For brevity, detailed plots for all devices are not included; however, each interferometer produced traces and fitted curves analogous to those in Fig. 6 and was processed with the same workflow.

TABLE II
COMPARISON OF FSR AND GROUP INDEX VALUES AFTER MEASUREMENTS FOR DIFFERENT ΔL.

	Simulation		Measured Results	
ΔL [μm]	FSR [nm]	n_g calc.	FSR [nm]	n_g est.
100	5.780	4.176	5.7554	4.1901
120	4.807	4.179	4.7909	4.1947
600	0.956	4.193	0.9550	4.1960
800	0.717	4.193	0.7162	4.1942
1000	0.574	4.194	0.5729	4.1934
2000	0.287	4.194	0.2864	4.1923
3000	0.191	4.195	0.1909	4.1905

The experimentally extracted group index values fall within the range predicted by the corner analysis in Figure 4, namely

$$n_g \in [4.1606,\ 4.2398],$$

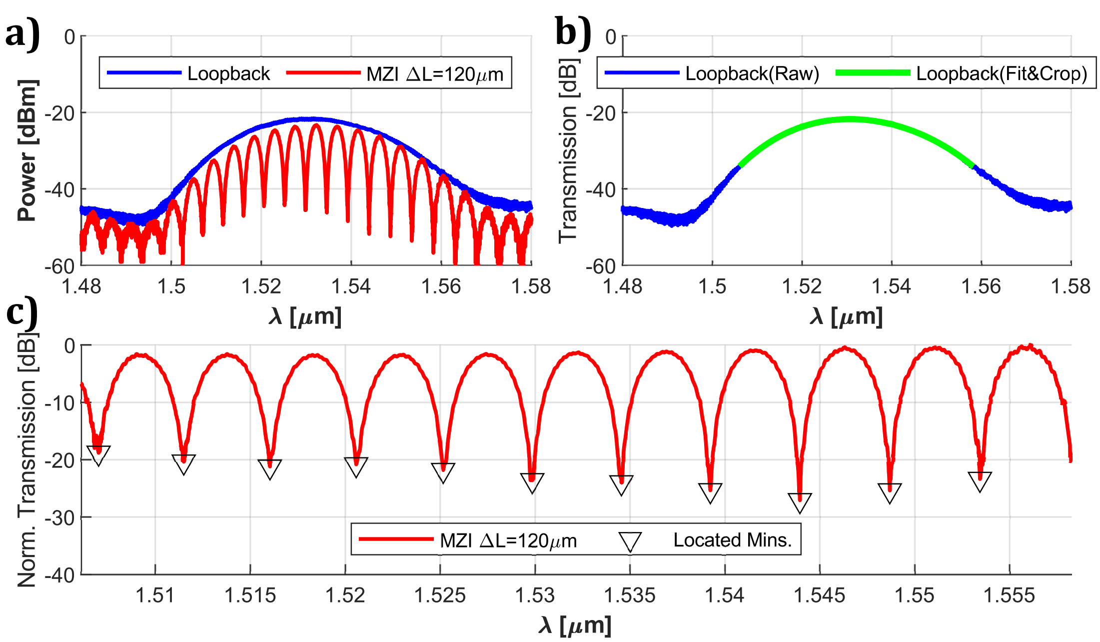

Fig. 5. Experimental results for the unbalanced MZI with $\Delta L = 120\ \mu$m. (a) Raw loopback measurement used to capture the wavelength-dependent response of the grating couplers. (b) Loopback spectral envelope obtained through smoothing, used to normalize the MZI trace. (c) Normalized MZI transmission showing the interferometric pattern after removing the loopback contribution, with the located minima marked for FSR extraction.

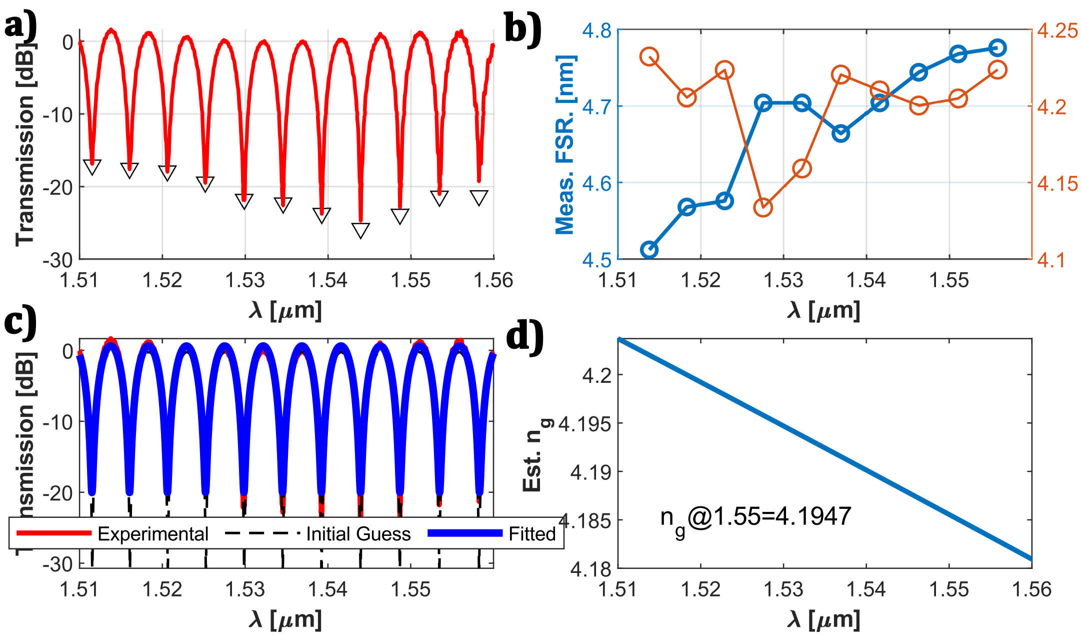

Fig. 6. Analysis workflow for the unbalanced MZI with $\Delta L = 120$ μm. (a) Normalized experimental spectrum after loopback removal. (b) Measured free-spectral-range (FSR) as a function of wavelength, extracted from the spacing between minima. (c) Fitted MZI transfer function overlaid on the measured data. (d) Estimated group index as a function of wavelength, with the retrieved value at 1.55 μm indicated.

consistent with the expected geometric variations in the width and height of the fabricated waveguides. Particularly, in this case, it seems that the actual geometry of the waveguide is close to the initial design target, with a maximum error percentage of 1%. More analysis should be done to confirm this, however I ran out of time. Results shown on this report are still under review.

6. Conclusions

The results presented in this report correspond to the current stage of the project, with data processing and analysis completed up to December 1st. The extraction of the group index from unbalanced MZIs has been successfully demonstrated for all fabricated path-length differences, and the measured values fall within the range predicted by the corner analysis. The implemented processing workflow—loopback normalization, FSR extraction, and transfer-function fitting—has provided consistent results across devices.

During the analysis, it was observed that for the largest path-length differences, the quality of the nonlinear fit tends to degrade. This appears to be related to the increased sensitivity of the model to local variations in the FSR estimation. As future work, an initial estimate of the dispersion could be incorporated into the fitting routine, allowing the model to better capture wavelength-dependent behaviour and potentially improve the robustness of the parameter extraction for long-delay interferometers.

Overall, the methodology is functioning, and the current results establish a baseline for the next steps.

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